MIL-P-82442(OS)
10 May 1966
Superseding
WS-8224
4 January 1966

MILITARY SPECIFICATION

PUMP, SEA WATER

This Specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of vane pump referenced herein as the "pump".

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military	
MIL-F-116	Preservation, Methods of
MIL-G-4343	Grease,Pneumatic System
MIL-F-18866	Fittings, Flared Tube, 37-Degree Steel, (Three Piece)
MIL-G-238N	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-S-45180	Sealing Compound, Gasket Hydrocarbon Fluid, Water Resistant

STANDARDS

Military	
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS-15795	Masher', Flat, Metal, Round General Purpose
MS-16555	Pins, Straight Headless Dowell, 001 Over Nan Size
MS-29561	Packing, Preformed, O, Ring, Synthetic Lubricant Resistant
MS-35307	Screw Cap, Hexagon, Head Finished Hexagon, Bolt, Corrosion. Resisting Steel Passivated, UNC-2A
MS-35678	Pin, Grooved Headless, Longitudinal Straight Gorrve, Corrosion-Resisting Steel
AN-932	Plug, Countersink Hex Head Pipe

DRAWINGS

Bureau of Naval Weapons	
DL-1568923	Pump Assembly, Sea Water
CD-1557052	Packing, Preformed
CD-1568913	Rotor, Pump
CD-1568914	Pressure Plate, Pump
CD-1568917	Pump, Body Assembly

FSC-1355

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CD-1568923	Pump Assembly, Sea Water
CD-2526581	Vine, Large
CD-2526637	Spacer, Pin
CD-2526735	bearing, Needle
CD-2526737	Wear Plate, Pump

(Copies of specifications, standard, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

National Aircraft Standards

NAS-617	Packing, Preformed

(National Aircraft Standards maybe obtained from National Standards Association, inc. 1321 Fourteenth St. N. U. Washington, D.C. 20005)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the pump l re classified as follows:

REQUIREMENTS	PARAGRAPH
General Requirements	3.2
Performance Requirements l nd Product Characteristic	3.3
Environmental Requirements	3.4
Operational Semite Life	3.5

3.2 General requirements.

3.2.1 Data requirements. No data is required by this l pacification or approval documents referenced in Section 2 unless specified in contract or purchase order. See 6.2.

3.2.2 Preproduction sample. Unless otherwise specified in the contract or purchase order , the preproduction sample of the pump shall be manufactured using the methods and procedures proposed for production. The sample will be tested l s specified in Section 4 herein l nd is for the purpose Of determining that prior to starting production, the contractor's production methods l re capable of yielding items chat comply with the technical requirements of the contract.

3.2.3 Materials. Unless otherwise specified in the contract or purchase order, l ll materials l nd treatments of materials l nd parts used in the manufacture of the pump shall be in l ccordance with the l pplicable documents listed on BuWeps DL 1568923.

3.2.4 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and product characteristics. Unless specified otherwise the pumps shall satisfy the following performance requirements when tested l s specified in Section 4.

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3.3.1 Pressure and flow requirements. The pump shall provide a discharge at a flow rate of 8.3 gallons per minute (gpm) minimum, l t a pressure of 265 pounds per square inch gage (psig) minimum when operated at 3880 ± 10 revolutions per minute (rpm), with l n inlet pressure of 15 psig.

3.3.2 Static case pressure. The pump shall be capable of withstanding internal pressures of 1250 psig minimum without leakage.

3*3*3 Fluid. The pump shall be capable of pumping fresh water or a salt solution (6.3.9} l t temperatrues between +32° F and +98° F.

3.3.4 Duty cycle. The pump shall be capable of operation for 10 duty cycles of 7 minutes each, at rated discharges as specified in 3.3.1

3.4 Environmental requirements. Unless otherwise specified, the pump shall be capable of conforming to the performance requirements and product characteristics of the specification l fter being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and non-operating see 6.3.1, 6.3.2, l nd 6.3.3.

3.4.1 Vibration, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meting the performance requirements l fter being subjected, in the ready condition, to the vibrations defined in Table X. The pump shall be vibrated for 180 minutes l ach in its longitudinal l xis and in two mutually perpendicular transverse axes l t any temperature within the range of -45" F to +140° F. Vibration is defined in 6.3.4. "

TABLE I

VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (cps)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

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3.4.2 Acceleration, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table II for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

TABLE II
ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III. Shock is defined in 6.3.6.

TABLE XII
SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction Of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity.

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3.4.4.1 Non-operating.

a. Temperature. The pump shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

b. Humidity The pump shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

3.4.4.2 Ready, operating. The pump immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F shall be capable of meeting the performance requirements, using a salt solution, at any temperature between +32°F and +98° F.

3.5 " Operational service life. The pump shall have a minimum service life of 70 minutes which shall be divided into 10 duty cycles of 7 minutes each.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade–to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Governments's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless othewise specified in the contract or purchase order, the supplier Is responsible for the performance of aLl inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection of the pump shall be classified as follows:

a. Preproduction inspection (Subsection 4.4)
b. Acceptance inspection (Subsection 4.5)
c. Periodic inspection (Subsection 4.6)
d. Examination of first echelon repair parts (Subsection 4.7)

4.3 Sampling.

4.3.1 Preproduction inspection sample. A preproduction sample of 3 pumps manufactured in accordance with 3.2 shall be subjected to the preproduction tests listed in Table IV and detailed in 4.4 through 4.4.4.10 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the pump by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

4.3.2 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is fount.

when a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspect ion is resumed.

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4.3.3 Examination of first echelon repair parts. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of first echelon repair parts for the pump as specified in 4.7 through 4.7.4.

4.4 Preproduction inspection.

4.4.1 Tests and examinations. The preproduction sample of 3 pumps shall be subjected to the acceptance tests defined in 4.5.4.1 and 4.5.4.2 and subsequently shall be inspected in accordance with Table IV, with 1 pump being tested in accordance with each of the 3 test sequences, A through C.

TABLE IV

PREPRODUCTION TEST SEQUENCE

Tests	Paragraph Reference	Test Sequence A	B	C
Operational Test	4.4.4.1	1	1	1
Vibration, Ready Condition	4.4.4.2	2	2	
Acceleration, Ready Condition	4.4.4.3	3		
Shock, Ready Condition	4.4.4.4	4		
Operational Test	4.4.4.1	5	3	
Temperature and Humidity	4.4.4.5	6	4	
Low Temperature Operational Test	4.4.4.6	7		
High Temperature Operational Test	4.4.4.7		5	
Vibration, Shipping	4.4.4.8			2
Shock, Shipping	4.4.4.9			3
Operational Service Life	4.4.4.10	8	6	4
Pump "A" Seven operational tests				
"B" Seven operational tests				
"C" Nine operation tests				

4.4.2 Test conditions. Unless otherwise specified, the pump shall be subjected to the tests in 4.4.1 under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.4.3 Test equipment and set-up. In addition to the items listed in 4,5,3, test equipment is required which provides the following characteristics:

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4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep 5 to 500 cycles per second (cps)
b. Test specimen: 5 pounds
c. Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	10 g's

d. Acceleration (packed and packaged):

5 to 60 cps	1 g

e. Automatic cycling time range: 0 to 15 minutes (optional)
f. Temperature chamber range: -45" F to +140° F accuracy to ± 5° F
g. Interior size. 7-1/2 inches x 6 inches x 8 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 5 pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 5 pound test specimen and shall provide ranges of:

a. 300 g's for 0.5 millisecond (ms) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics.

a. Temperature: -45° F to +140° F; with an accuracy to ± 5° F
b. Relative humidity: 95 percent minimum at +110° F; with an accuracy to ± 5° F
c. Interior size: 7-1/2 inches x 6 inches x 8 inches minimum

This equipment may be used for both non-operating and ready/operating temperature (and humidity) tests.

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4.4.3.5 Salt water supply. Salt water (6.3.5) shall be used as the coolant . The salt water supply shall be capable of providing salt solution at a rate of 10 gpm at temperatures between +32' F and +98° F at a pressure of 15 psig, measured at inlet test port.

4.4.3.6 Temperature indicator. Range: 0° F to +140° F, accuracy to = 2° F.

4.4.4 Test Procedures .

4.4.4.1 Operational test. Install the pump in the pump test stand and connect to appropriate fittings in accordance with Figures 1 and 2. With the drive unit operating the pump at 3880 ± 10 rpm, the output pressure shall be: 265: psig minimum. The pump shall operate for one complete duty cycle of 7 minutes on salt water. Post-test servicing and maintenance shall be performed in accordance with 4.5.7.

4.4.4.2 Vibration. ready condition. The pump, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The pump shall then be subjected to resonance and cycling vibration in each axis , in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ± 5° F, room temperature and +140° F ± 5° F respectively, For cycling, a logarithmic frequency sweep rate shall be used if available; othewise, other cycling rates of frequency change may be used. The sweep-rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance *	-	30 min	1 hr	1-1/2 hr	2 hr
Cycling Time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

* 30 minutes each resonance

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TABLE VI

VIBRATION SCHEDULE

Frequency (cps)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2		0.036	
Acceleration (g's ± 10%) Minimum		3		10

4.4.4.3 Acceleration, ready condition. The pump, in the ready condition, shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the pump.

TABLE VII

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.4 Shock, ready condition. The pump, in the ready condition, shall be attached to a shock test machine (see Figure 1). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the pump shall be performed in accordance with Table VIII in the sequence shown.

TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	-	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

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4.4.4.5 Temperature and humidity , non-operating condition. Two preproduction pumps shall be subjected, in the non-operating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F + 0° - 5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° F + 0° -5° F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours) . The specimen shall then be subjected to an environmental temperature of +140° F + 5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F +5° -0° F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, one pump shall be operated in accordance with 4.4.4.6 and one pump shall be operated in accordance with 4.4.4.7. Prior to operation the pump may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.6 Low temperature operating. Before testing, the pump, installed in its test chamber in the ready condition, shall be conditioned at a temperature of 0° F ± 5° F for a period of 1 hour. Mediately thereafter the pump shall be operated in accordance with and meet the requirements of 4.4.4.1 at an environmental temperature of +32° F ± 5° F.

4.4.4.7 High temperature operating. Before testing, the pump, installed in its test chamber in the ready condition, shall be conditioned at a temperature of +140° F ± 5° F for a period of 1 hour. Immediately thereafter the pump shall be operated in accordance with and meet the requirements of 4.4.4.1 at an environmental temperature of +140° F ± 5° F.

4.4.4.8. Vibration, packed and packaged. The pump, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.

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4.4 .4.9 Shock, packed and packaged . The pump, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The pump in its container shall be dropped from a height of 36 inches, one drop on each of 4 diagonally opposite corners and flat drops, one each, on the bottom and on one side. The drops shall be made on an impact surface of essentially non-yielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -45° F ± 5° F, and within 10 minutes after the container has been stabilized at +140° F ± 5° F.

4.4.4.10 Operational service life. Each preproduction sample shall complete the life test requirement by being subjected to a total of ten 7-minute operational tests in accordance with 4.4,4.1 during the pre-production test schedule, Table IV. The number of additional operational tests needed to meet the life requirement is listed in Table IV under "Operation Service Life".

4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier and maintained as part of the inspection records. In addition to the data requirements of 4.5.5, the data sheets shall contain, but are not limited to, the following information:

a. Low temperature operating test: Pump conditioned for: ____hrs. at ____° F Salt Solution, Temperature ____° F

b. High temperature operating test: Pump conditioned for: ____hrs. at ____° F Salt Solution, Temperature ____° F

4.4.6 Determination of defect or failure. Failure of a pump to successfully pass any of the tests in 4.4 through 4.4.4.10 shall be cause for rejection of the preproduction sample.

4.5 Acceptance inspection. Each pump produced under the contract or purchase order shall be subjected to the following tests and examinations.

4.5.1 Tests and examinations.

a. Static pressure test
b. Operation

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4.5.2 Test conditions. Unless otherwise specified, the pump shall be subjected to the acceptance tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.5.3 Test equipment and set-up. The pump test stand shall provide fixtures and accessories for mounting the pump, a fluid supply, a drive unit and pressure gages. The following items of equipment are required and shall have the characteristics indicated.

a. Pressure gage:
Range: 0 to 400 psig, accuracy to ± 2 percent full scale
Operating temperature: +25° F to +100° F minimum
Gage size: 6-inch minimum

b. Drive unit:
Horsepower: two nominal
Speed: 3880 ± 10 rpm
Operating temperature: +25° F to +120° F minimum

c. Water supply:
The water supply shall be capable of providing tap water at a rate of 10 gpm at temperatures between +32° F and +98° F at a pressure of 15 ± 2 psig at the pump inlet.

d. Relief valve:
Cracking pressure: 50 psig
Closing pressure: 30 psig

e. Fixed orifice:
When connected in the test configuration the fixed orifice shall be capable of producing a pressure of 275 ± 10 psig at gage A at a flow rate of 8.3 gpm.

4.5.4 Test procedures.

4.5.4.1 Pressure test. Plug the inlet port, plug one of the outlet ports, and attach an adaptor to the other outlet port. See Figure 1. Pressurize with hydraulic fluid to 1250 psig for five minutes. There shall be no evidence of continuous leakage.

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4.5.4.2 Operation. Install the pump in the pump test stand and connect the applicable fittings in accordance with Figures 1 and 2. With the pump operating at 3880 ± 10 rpm, inlet pressure of 15 ± 2 psig, the output pressure shall be: 265 psig minimum. The pump shall be operated for 1.5 minutes on tap water. Post-test servicing and maintenance shall be performed in accordance with 4.5.7.

4.5.5 Test data sheets. All test data shall be recorded on data sheets by the supplier. Data sheets shall contain, but are not limited to, the following information:

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a. Date _____
b. Pump serial number _____
c. Date of instrumentation calibration _____
d. Inlet pressure _____psig
e. Outlet pressure test data _____ psig _____minutes
f. Speed _____ rpm
g. Post-test servicing by _____

4.5.6 Determination of defect or failure. Failure of a pump to successfully pass the tests of 4.5.4.1 and 4.5.4.2 shall be cause for rejection of that pump.

4.5.7 Pose-test servicing and maintenance. The pump shall be disassembled and cleaned after each run. All packing preforms shall be inspected and replaced if required. The pump shall then be reassembled.

4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.2. The samples shall be tested in accordance with the following plan:

a. The Sample of 1 unit from the first group of 30 units will be called the first Sample and will undergo the tests of Table IX for Sample 1. The sequence of tests will be as shown in Table XX.

b. The Sample of 1 unit from the second and third groups of 30 units will be called the second and third Samples and will undergo the tests of Table IX for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table IX.

c. The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This sample will be called the fourth sample and will undergo the tests of Table IX for Sample 4, etc.

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TABLE IX

PERIODIC TEST SCHEDULE

	Sample and Test Sequence		
	1, 4, Etc.	2, 5, Etc.	3, 6, Etc.
Vibration, Ready Condition		1*	1*
Acceleration, Ready Condition			2*
Shock, Ready Condition	1*		3*
Temperature and Humidity		2	4
High Temperature Operating		3	5
Low Temperature Operating	2		6
Operational Test	3	4	
Operational Service Life			7

* Visual Check After Testing

4.6.1 Test conditions. Unless Otherwise specified, the pump shall be subjected to the periodic tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98°F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.6.2 Test equipment and set-up. The test equipment and set-up for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures.

4.6.3.1 Vibration, ready condition. Vibration testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.2.

4.6.3.2 Acceleration, ready condition. Acceleration testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.3.

4.6.3.3 Shock, ready condition. Shock testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.4.

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4.6.3.4 Temperature and humidity, non-operating condition. Temperature and humidity testing of the pump, in the non-operating condition, shall be performed in accordance with 4.4.4.5.

4.6.3.5 High temperature operating. The pump shall be subjected to the high temperature operating test in accordance with 4.4.4.7.

4.6.3.6 Low temperature operating. The pump shall be subjected to the low temperature operating test in accordance with 4.4.4.6.

4.6.3.7 Operational test. The pump shall be subjected to the operational test in accordance with 4.4.4.1.

4.6.3.8 Operational service life. The pump shall be subjected to 8 additional operational tests in accordance with 4.4.4.1

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of a pump to successfully pass the tests in 4.6.3.1 through 4.6.3.8 shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.2.

4.7 Examination of first echelon repair parts.

4.7.1 General.

4.7.1.1 The information presented in this section is applicable to all components assembled into the end product and to first echelon repair parts. The information is divided into the following major divisions:

a. Parts covered by separate CD's
b. Parts which are manufactured in accordance with Federal or Military specifications which contain adequate quality assurance provisions.
c. Minor hardware items

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4.7.1.2 Unless otherwise specified, sampling plans and procedures set forth in MIL-STD-105, Inspection Level 11 shall be used in determination of acceptability of the first echelon repair parts and all components assembled into the end product. An AQL of 1.0 shall be used for major items of inspection, an AQL of 2.5 shall be used for minor items of inspection.

4.7.1.3 The verification of the quality of repair parts is mandatory and shall be performed whether the item is a part of a repair kit or is to be assembled into the end product.

4.7.2 Separate Classification of Defects (CD's) The components and assemblies referenced within this subsection shall be inspected in accordance with the applicable CD's. For inspection of repair parts use:

	SPARE PARTS	BU WEPS CD
	Packing, Preformed	1557052
	Rotor, Pump	1568913
	Pressure Plate, Pump	1568914
*	pump Body Assembly	1568917
	Pump Assembly, Sea Water	1568923
	Vane, Large	2526581
	Spacer, Pin	2526637
	Bearing, Needle	2526735
	Wear Plate, Pump	2526737

* The static pressure test shall be waived for those items that are components of sea water pump assemblies and tested in accordance with 4.5.4.1.

4.7.3 Components covered by Federal and Military Specifications. The components and parts referenced within this subsection shall be inspected in accordance with the applicable specifications. For inspection of repair parts use:

PART	SPECIFICATION
Grease	MIL-G-3278
Lubricant	MIL-L-4343
Elbow 90°	MIL-F-18866
Sealer	MIL-S-45180

4.7.4 Miscellaneous hardware. The items referenced within this subsection shall be inspected as to type and quantity.

PART	IDENTIFICATION
Washer, Flat	MS- 15795-810
Pin	MS- 16555-640
Packing, Preformed	MS- 29561-012,014,016
Screw, Cap, Hex Hd.	MS- 35307-308

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PART	IDENTIFICATION
Pin, Groove	MS- 35675-10
Plug, Csk Hex Hd Pipe	AN- 932-S2, S3
Packing, Preformed	NAS- 617-4

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified. See 6.2. Unit package shall consist of one pump. l

5.1.1 Level A. Preservation and packaging of equipment for level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished with Method D-1, or D-4. Preservation and packaging shall be Method IA.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be accomplished in accordance with the supplier's own commercial practice when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be level A or C as specified. See 6.2. Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with MIL-P-116, Method 11 D.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The equipment, when preserved and packaged in accordante with 5.1.3, shall be overpacked in compliance with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

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5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect the pump from damage due to shock and vibration loads encountered during all modes of handling and transportation. Shock and vibration tests shall be in accordance with this specification.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping, containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The sea water pump is intended for use in a torpedo power plant to supply sea water, as a coolant, to an engine, combustion chamber, and alternator,

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required
e. Required level of preservation, packaging, and packing

6.3 Definitions.

6.3.1 Ready. The condition of the pump when fully assembled, after completion of servicing and maintenance.

6.3.2 Operating. The condition of the pump when operating.

6.3.3 Non-operating. The condition of the pump when stored, transported, or handled. Tests on the pump, in the non-operating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in cycles per second.

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6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 seconds. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up to 90 percent of peak force in less than 0.010 seconds. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form equivalent to one half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock waveform builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the pump is exposed, is expressed in degrees Fahrenheit.

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the pump is exposed, expressed in percent.

6.3.9 Salt solution. The salt solution shall be prepared by dissolving 3 parts, by weight,of sodium chloride in 97 parts, by weight, of tap water.

6.4 Weight. The weight of the pump shall not exceed 2 pounds.

6.5 Storage life.

6.5.1 Packed and packaged condition. The pump shall be capable of with-standing storage for 5 years in the packed and packaged condition and, thereafter, meet the performance requirements of this specification.

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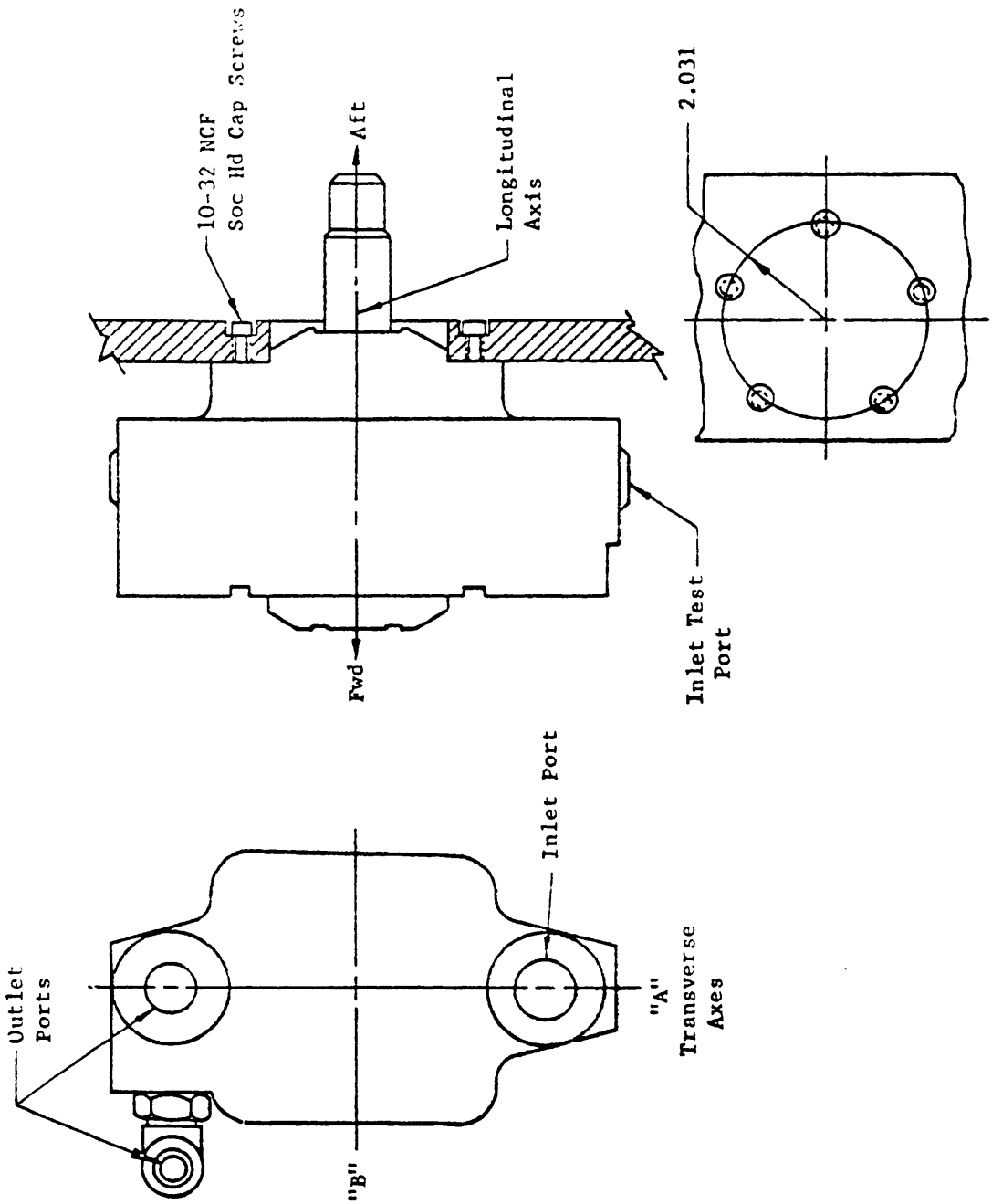


Figure 1, Sea Water Pump Mounting

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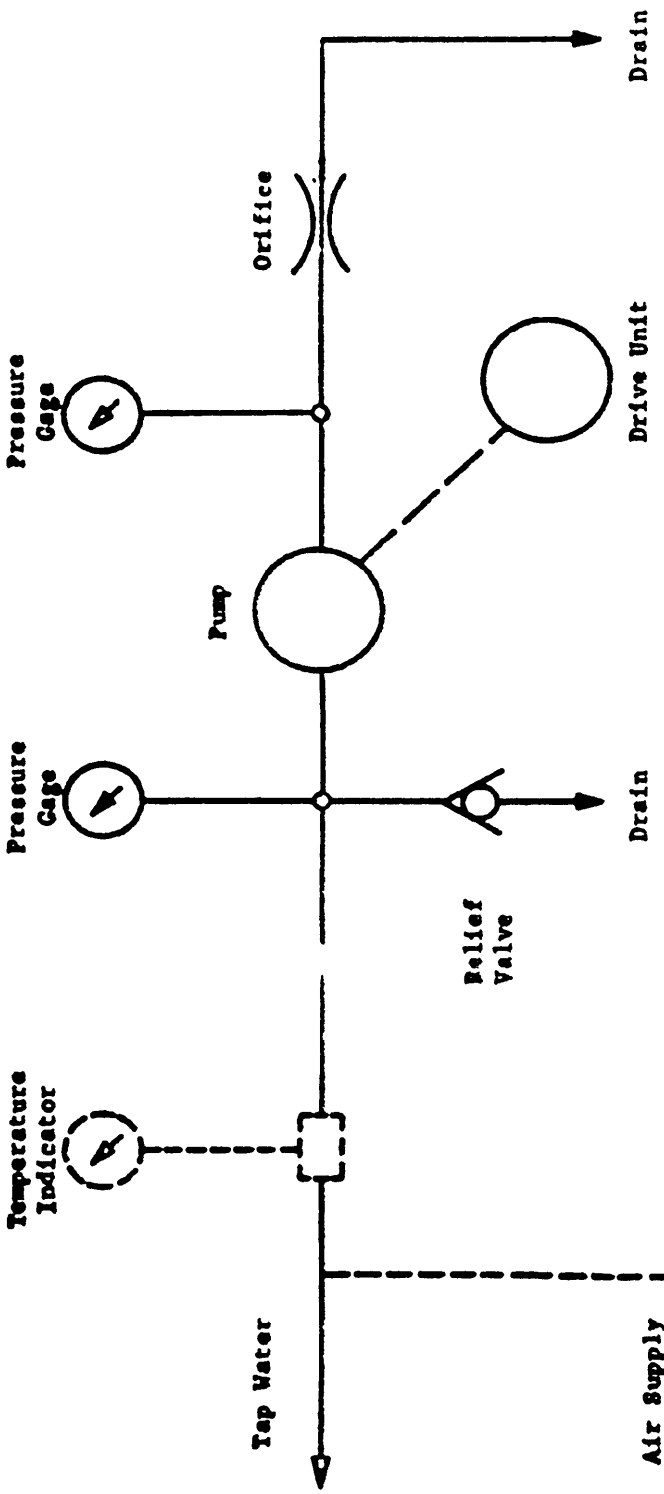


Figure 2, Sea Water Pump Flow Diagram

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-P-82442(OS) PUMP, SEA WATER

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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